

1. Logical Operators (and, or, not) Questions

Question 1:

Python

Take age and marks from user.

Check if the student is eligible for admission:

Age must be between 18 and 25 AND marks >= 60

Question 2:

Python

Take username and password from user.

Check login:

Username == "admin" AND Password == "12345"

Print "Login Successful" or "Login Failed"

Question 3:

Python

Take a number from user.

Check if the number is between 50 and 100 (using 'and')

Also check if it is NOT divisible by 5 (using 'not')

Question 4:

Python

Take three numbers from user.

Print the largest number using 'and' and 'or' operators.

2. String and String Slicing Questions

Question 5:

Python

```
# Take your full name as input from user.  
# Print the following:  
# 1. Full name in uppercase  
# 2. Full name in lowercase  
# 3. Length of your name  
# 4. First 3 characters  
# 5. Last 3 characters
```

Question 6:

Python

```
# Take any sentence from user.  
# Print:  
# - First 10 characters  
# - Last 10 characters  
# - Characters from index 5 to 15  
# - Every 2nd character of the sentence
```

Question 7:

Python

```
# Take a word from user.  
# Check if the word is a palindrome or not.  
# (Example: "madam", "radar")
```

Question 8:

Python

Take email as input (e.g: student@gmail.com)

Print only the username (part before @)

Print only the domain (part after @)

Bonus Mixed Question (String + Logical)

Question 9:

Python

Take a password from user.

Check if password is strong:

- Length ≥ 8

- Contains '@' or '#'

Print "Strong Password" or "Weak Password"

1. Logical Operators

Python

Take three subject marks from user (English, Math, Science).

Check if the student has passed:

All marks should be ≥ 40 (use 'and' operator)

If any one subject is < 40 , print "Fail"

2. Logical Operators + User Input

Python

Take a number from user.

Check whether the number is:

```
# Positive AND Even
# OR
# Negative OR Odd
# Print appropriate message
```

3. String Operations

Python

```
# Take your city name as input.
# Print the following:
# 1. City name in reverse order
# 2. Number of vowels in the city name (a,e,i,o,u)
# 3. City name without first and last character
```

4. String Slicing

Python

```
# Take a mobile number as input (10 digits).
# Print:
# First 3 digits (network code)
# Last 4 digits
# Middle 3 digits
```

5. Mixed (String + Logical)

Python

```
# Take a string from user.
# Check if:
# - The string length is greater than 5
```

- AND it starts with 'A' or 'a'

Print "Valid String" or "Invalid String"